

January 2023 Exam

ST443

Machine Learning and Data Mining

2020/21 and 2021/22 syllabus - resit/deferred candidates only

Instructions to candidates

This paper contains five questions. Answer **ALL Five**.

Question 1: 19 marks

Question 2: 19 marks

Question 3: 22 marks

Question 4: 20 marks

Question 5: 20 marks

The marks in brackets reflect marks for each part of a question.

Time allowed **Reading Time:** *None*
 Writing Time: *2 hours*

You are supplied with: *No additional materials*

You may also use: *No additional materials*

Calculators: *Calculators are allowed in this exam*

1. (a) Select *True* or *False* for the following statements. Briefly justify your answers.
 - i. Both k -nearest neighbours and logistic regression are nonparametric classification methods. [2 marks]
 - ii. If we train a linear regression estimator on only half the data, the variance of the estimator will be larger than training it on the entire dataset. [2 marks]
 - iii. For the coefficient vector of multivariate linear regression with i.i.d. errors, the maximum likelihood estimator is always the same as the least squares estimator. [2 marks]
 - iv. Support vector classifiers directly give us the posterior probabilities $P(Y = 1|X)$ and $P(Y = -1|X)$. [2 marks]
 - (b) A natural degree- d spline is degree- d spline with additional boundary constraints: the function is required to be linear at the boundary (i.e. in the region where X is smaller than the smallest knot, or larger than the largest knot). How many degrees of freedom as a function of K does a natural degree-5 spline with K ($K > 2$) knots have? Justify your answer. [6 marks]
 - (c) Consider applying the bagging technique to the binary classification tree. Provide an example that the majority vote approach and the averaging probability approach can give you different classification results. [5 marks]
2. (a) Assume that we take a dataset, divide it into equally-sized training and test sets, and then try out two different classification procedures. First we use linear discriminant analysis and get an error rate of 20% on the training data and 15% on the test data. Next we use 1-nearest neighbours (i.e. $k = 1$) and get an average error rate (averaged over both test and training datasets) of 10%. Based on these results, which method should we prefer to use for classification of new observations? Why? [5 marks]
 - (b) Consider the labeled dataset in Figure 1. Is it possible for a k -nearest neighbours classifier to assign the negative class to the query point '?'? If so, what is the minimal value of k for which this occurs? Explain your answer. [6 marks]

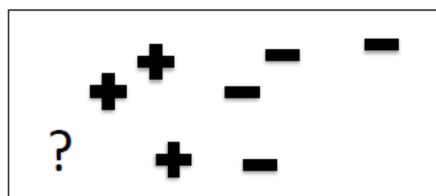


Figure 1: For Question 2. (b)

- (c) Provide advantages and disadvantages of the k -fold cross-validation relative to:
 - i. The validation set approach

- ii. Leave-One-Out-Cross-Validation
- iii. The bootstrap procedure.

[8 marks]

3. Suppose $\mathbf{Y} \in \mathbb{R}^d$ is a random vector distributed as $\mathcal{N}(\boldsymbol{\theta}, \mathbf{I}_{d \times d})$ for some unknown parameter $\boldsymbol{\theta} \in \mathbb{R}^d$. We observed a sample $\mathbf{y} \in \mathbb{R}^d$ of $\mathbf{Y}$ and want to estimate $\boldsymbol{\theta}$.

(a) What is the maximum likelihood estimator (MLE) of $\boldsymbol{\theta}$? [3 marks]

(b) Now we are going to use ridge regression to solve this problem. Namely, solve

$$\min_{\boldsymbol{\theta}} \left\{ \|\mathbf{y} - \boldsymbol{\theta}\|^2 + \frac{1}{2} \lambda \|\boldsymbol{\theta}\|^2 \right\}$$

to get an estimate of $\boldsymbol{\theta}$. What is the closed form of estimator $\hat{\boldsymbol{\theta}}(\mathbf{y})$ from ridge regression with regularization parameter $\lambda \geq 0$? [4 marks]

(c) Derive the population risk $\mathbb{E}\|\mathbf{Y} - \hat{\boldsymbol{\theta}}(\mathbf{y})\|^2$ for ridge regression estimator, where $\mathbf{Y}$ is from test data and is independent of the training sample $\mathbf{y}$. [5 marks]

(d) What is the population risk for the MLE estimator? [4 marks]

(e) Suppose we choose $\lambda = d$, find out the condition on $\boldsymbol{\theta}$ such that the ridge regression estimator has a lower risk than the MLE estimator. [3 marks]

(f) This implies that MLE estimator, although seems to be the most natural estimator, does not always achieve the lowest risk. Briefly explain the reasons behind this fact. [3 marks]

4. (a) We have the data in Figure 2 to learn a classification tree which predicts if people pass the machine learning course (Yes or No), based on their previous GPA (High, Medium, or Low) and whether or not they studied. Each split can be made based on the cost function

$$C(T) = \sum_{m=1}^{|T|} N_m Q_m,$$

where the cross-entropy $Q_m = -\sum_k \hat{p}_{mk} \log_2(\hat{p}_{mk})$ is used as the node impurity measure.

For each split, we consider the following candidate splits:

- $R_1 = \{\text{GPA} = \text{L}\}$, $R_2 = \{\text{GPA} = \text{M}\}$ and $R_3 = \{\text{GPA} = \text{H}\}$.
- $R_1 = \{\text{Studied} = \text{F}\}$ and $R_2 = \{\text{Studied} = \text{T}\}$.

i. For the first split, compute $C(T)$ for splits in i. and ii. Among these two candidate splits, which attribute is chosen to make the first split? [8 marks]

ii. Draw the final classification tree. [5 marks]

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

Figure 2: The data for Question 4 (a)

- (b) Consider a binary classification problem, $Y = 1$ (class 1) and $Y = 2$ (class 2). Suppose that $X|Y = 1$ is from a normal distribution $\mathcal{N}(2, 1^2)$ and $X|Y = 2$ is from a normal distribution $\mathcal{N}(1, 0.5^2)$. In addition, $P(Y = 1) = 1/3$. Derive the Bayes decision rule that classifies regions to classes 1 and 2 for this problem. [7 marks]
5. (a) Suppose that we have five observed points, each with four features. We present the Euclidean distance between any two observations with measurements on these four features in the following matrix.

	1	2	3	4	5
1	0.00	0.90	0.36	0.45	0.60
2	0.90	0.00	0.55	0.50	0.04
3	0.36	0.55	0.00	0.57	0.20
4	0.45	0.50	0.57	0.00	0.30
5	0.60	0.04	0.20	0.30	0.00

Use the matrix with Euclidean distances to perform hierarchical clustering, using average linkage. [10 marks]

- (b) Let $x_1, \dots, x_n$ be n observations and $\{C_1, \dots, C_K\}$ be a partition of $\{1, \dots, n\}$. Let n_k be the number of observations in cluster C_k for $k = 1, \dots, K$. For each cluster C_k , define $\bar{x}_k = \frac{1}{n_k} \sum_{i \in C_k} x_i$ to be the within-cluster mean and $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ to be the overall mean. Let

$$T = \sum_{k=1}^K \sum_{i \in C_k} (x_i - \bar{x})^2 \text{ to be the total deviance to the overall mean,}$$

$$W = \sum_{k=1}^K \sum_{i \in C_k} (x_i - \bar{x}_k)^2 \text{ to be the within-cluster deviance to the cluster mean,}$$

$$B = \sum_{k=1}^K \sum_{i \in C_k} (x_k - \bar{x})^2 \text{ to be the between cluster deviance.}$$

i. Verify that $T = W + B$. [6 marks]

ii. Explain how T and B change during the course of the k -means algorithm. [4 marks]